

CMPE257

Final Report

Group 7 - The Coconuts

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1. Idea

1.1. Description

Lending is a practice that spans hundreds of years. On a high level, lending is when a person with spare money (lender) gives some of it to a person who needs money (borrower). Then, after a fixed amount of time, a borrower is expected to return the money to the lender, and pay extra on top of the borrowed amount (interest) as a fee for a service.

Since each borrower can potentially go bankrupt and fail to repay the borrowed sum, careful decision-making is crucial for the lender to ensure that they will not lose their money. The history of how lenders made decisions on whether to lend or reject a borrower consists of multiple periods.

Around the 1800s, relationships, personal knowledge, trade references and blacklists dominated the scene. The reputation-based system is the oldest method for lenders to assess the background of the borrowers. This approach has the benefit of deeper personal context between the lender and borrower that allows for more flexibility and trust in the equation. On the other side of the equation, this approach is not scalable, and it has a high barrier to entry for new borrowers.

Around the 1950s, statistical scorecards emerged, and Fair, Isaac and Company (FICO) pioneered the credit score that most of us are familiar with. These novel approaches drove away the decision-making from the personal relations towards data-driven decisions. On a high level, these systems relied on a set of financial indicators of a borrower, such as the number of delinquencies, or the current amount of debt, to assess the borrower on a scale between “unlikely to repay the loan” to

“highly likely to repay the loan.” While these systems provided a scalable and reproducible way to quickly assess someone’s creditworthiness, the early implementations were flawed. Key issues revolved around decision-making transparency, credit file data quality, and nationwide outcome consistency. In the 1970s, there were rules and regulations implemented to govern how lenders can make decisions within the Fair Credit Reporting Act (FCRA) and the Equal Credit Opportunity Act (ECOA).

As of today, the major principles of creditworthiness decision-making are: usage of only allowed data variables, strict data collection and sharing principles, decision interpretability and consistency. With that, we recently saw the emergence of machine learning (ML) models that promise to provide better credit ranking decisions.

1.2. Goals/Objectives

In our project, we are attempting to train an ML model to assess whether a business is likely or not to repay the loan. As the algorithm input, we have a variety of variables tied to a business's past financial indicators, such as number of trades in the past 90 days, maximum delinquency on the record, external risk estimate, and more. As for the algorithm output, we train the model to classify a business into the bad/good borrower category. Based on our experiments, we were able to achieve high accuracy on a hold-out validation set without overfitting using an XGBoost model. This approach can provide better lending decisions that benefit both lenders and borrowers.

To summarize, the problem we are solving is not new; we are trying to decide whether someone who is asking for a loan is likely or not to repay it. The problem is significant because it

helps lenders to avoid losing money by not lending to entities who are not likely to pay back, and helps borrowers avoid taking on debt they may not be in the best position to repay.

2. Work Developed

2.1. Adopted ML algorithms/techniques

Having tried a few different approaches to cleaning the data, we stuck to replacing the invalid values with NaN and allowed our model (XGBoost) to handle them implicitly. A little more research also confirmed that XGBoost doesn't require feature scaling either, so we forewent that step as well.

We then flipped our target variables so that the Bad borrower class was assigned as '1' after encoding, to be able to scale the weights properly (the function `scale_pos_weights` works better when the more important class has a positive value based on our research). We then conducted a hyperparameter search after our existing baseline in the previous iteration of results, along with sweeping probability thresholds for classifications instead of relying on the default 0.5 value. This allowed for improvements in results in our "Bad" borrowers class, specifically the Recall and F1-Score, without affecting the overall AUC.

Next, we used our Case Study as a reference and added monotonic constraints, to enforce more consistent relationships between our features and the predicted class of the borrower. We used the Spearman correlation coefficients to identify monotonicity and its directions, and removed those with an absolute value less than 0.1. Next, we retrained the model with the resultant monotonic constraint vector (the provision for which was present in XGBoost's functions without any additional building on our end) and the best set of

hyperparameters we previously found, resulting in our final version of the model.

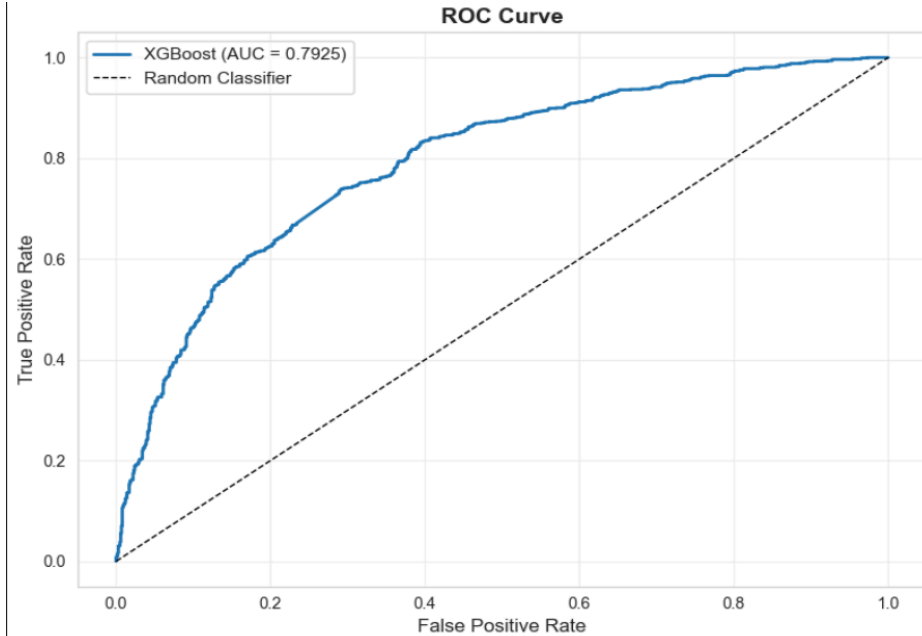
2.2. Performance metrics

The model’s predictive performance was evaluated using standard classification metrics appropriate for credit-risk modeling, where the objective is to reliably identify high-risk borrowers while limiting false positives. The model meets all predefined performance objectives, including minimum thresholds for ROC–AUC, recall, precision, and F1-score.

FINAL PERFORMANCE ASSESSMENT					
Objective	Target	Achieved	Status	Margin	
ROC-AUC $\geq$ 0.75	0.750	0.792479	✓ PASS	0.042479	
Recall (Bad) $\geq$ 0.70	0.700	0.840659	✓ PASS	0.140659	
Precision (Bad) $\geq$ 0.65	0.650	0.692308	✓ PASS	0.042308	
F1-Score $\geq$ 0.67	0.670	0.759305	✓ PASS	0.089305	
Beat Random Baseline (0.50 AUC)	0.500	0.792479	✓ PASS	0.292479	
Beat Majority Baseline (52.2% Acc)	0.522	0.721797	✓ PASS	0.199797	
OVERALL: 6/6 objectives met					

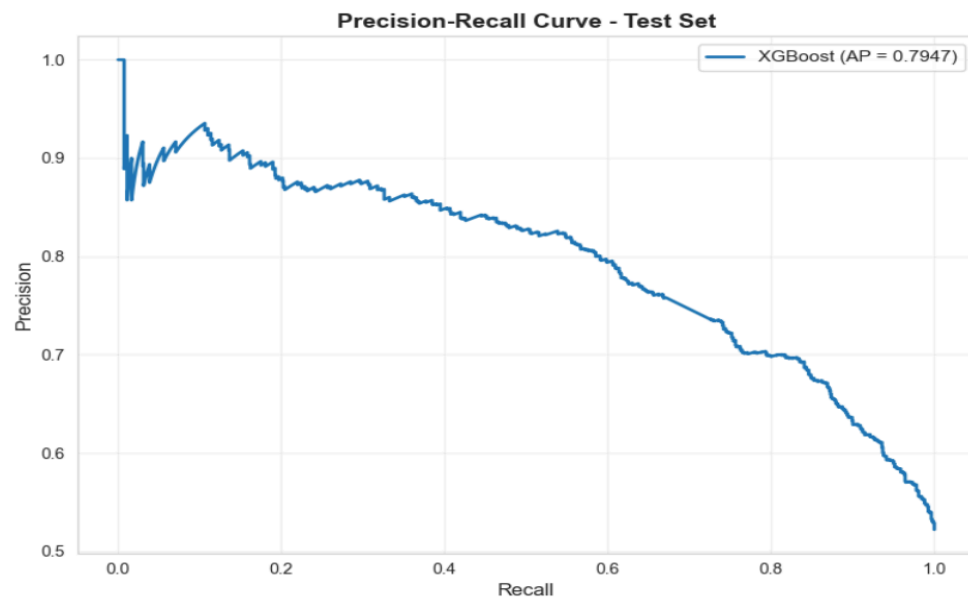
ROC–AUC measures the model’s ability to distinguish between Good and Bad borrowers across all thresholds.

The model achieves a ROC–AUC of 0.7925, which indicates strong discriminative performance for a tabular credit dataset.



Because the dataset is imbalanced and the goal emphasizes identifying Bad borrowers, the Precision–Recall curve provides additional insight.

The model obtains an Average Precision (AP) of 0.7947, indicating consistently strong precision across a wide range of recall values.



At the selected operating threshold of 0.40, the model achieves:

- Bad class recall: 0.8407
- Bad-class precision: 0.6923
- Bad class F1-score: 0.7593
- ROC–AUC: 0.7925

These numbers place the model's performance in the 0.79–0.80 AUC range, with Bad-class recall in the 0.84–0.85 range, consistent with the project's objective of prioritizing detection of high-risk borrowers.

Additionally, the incorporation of monotonic constraints slightly improved the Good-class recall while not reducing Bad-class performance, showing that interpretability constraints did not harm core predictive metrics.

2.3. Evaluation

We applied monotonicity constraints so the model would behave more like a real credit-scoring system.

For example, if someone has a higher credit score, the model shouldn't predict them as more risky.

The picture below shows the direction we enforced for each feature:

- +1 means the feature should *increase* risk
- –1 means it should *reduce* risk
- 0 means no constraint

This helps the model stay consistent with how real lenders think, and it makes the predictions easier to explain.

The picture(below) shows that most features end up following the expected direction, which makes the model easier to explain and justify.

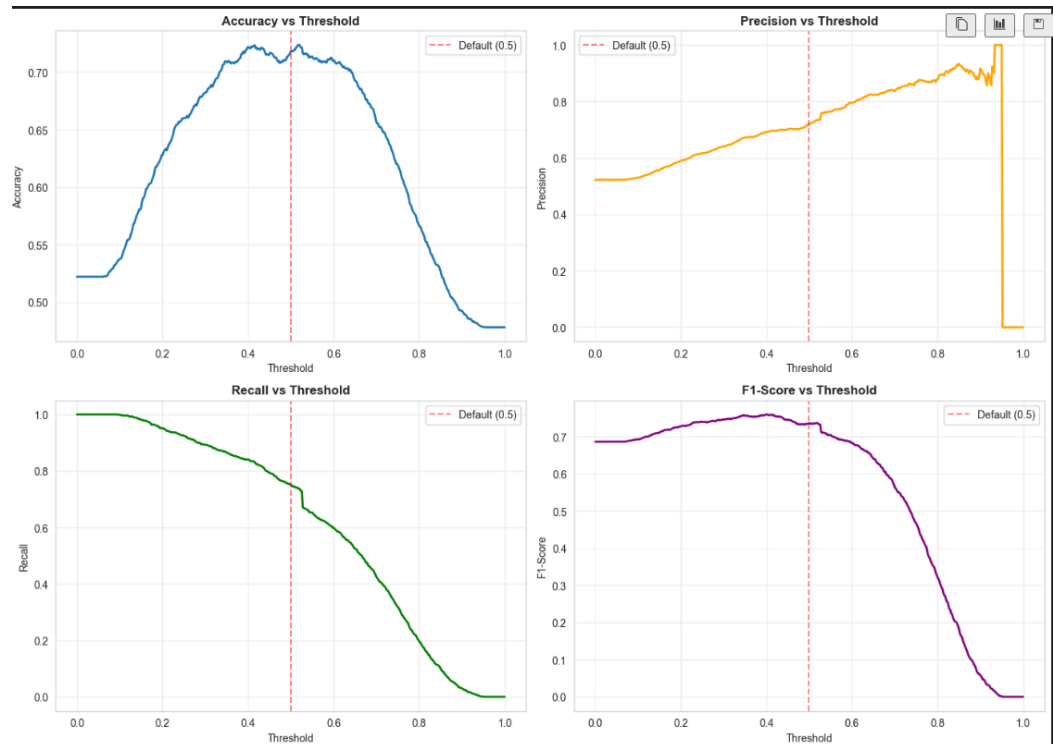
```
Monotonic Constraints Applied:  
Total features: 23  
Increasing (1): 9  
Decreasing (-1): 10  
No constraint (0): 4  
  
Constraints string: (-1,-1,0,-1,-1,1,1,-1,-1,-1,-1,-1,0,1,-1,1,1,1,0,1,0,1,1)
```

To choose the best classification threshold, we plotted how accuracy, precision, recall, and F1 change across different thresholds.

We found:

- F1-score peaks around 0.40
- Accuracy peaks around 0.52
- Recall drops quickly as the threshold gets higher
- Precision increases as the threshold gets higher

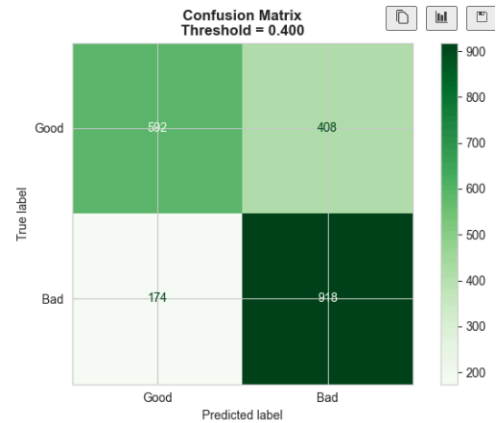
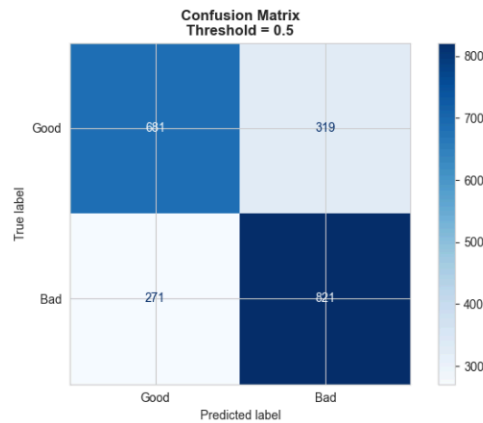
Since the project goal is to catch as many Bad borrowers as possible, we picked 0.40. This gives us a good balance between recall and precision.



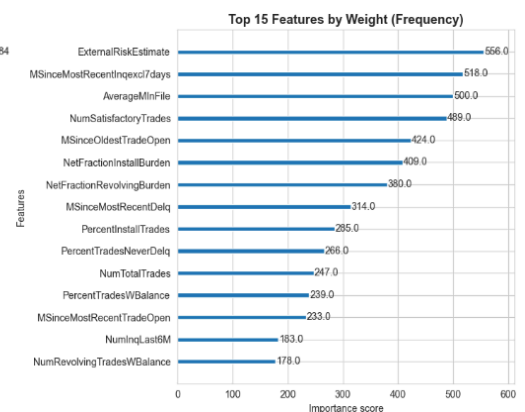
To see what the threshold actually changes, we compared confusion matrices for 0.50 and 0.40.

- At 0.50, the model misses 271 Bad borrowers.
- At 0.40, it misses 174, which is a big improvement.
- The cost is more false positives, meaning some Good borrowers get labeled as Bad.

For a lender, it's usually better to be safe and avoid lending to risky customers, even if that means rejecting a few extra good ones. So the lower threshold makes sense.

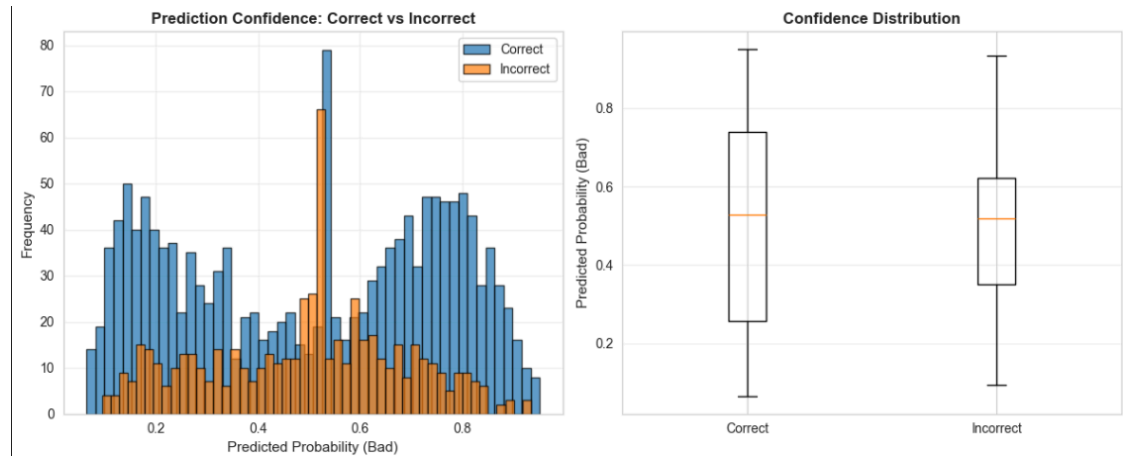


The model's most important features are things like ExternalRiskEstimate, PercentTradesNeverDelq, and revolving-burden variables. These match what you'd expect in a credit-risk context, which gives more confidence that the model is using reasonable information.



We also checked how the model's confidence changes between correct and incorrect predictions:

- Correct predictions usually have higher confidence
- Incorrect predictions mostly happen when the probability is near the threshold (around 0.35–0.55)
- This is typical for credit data, since a lot of applicants fall into a “gray area”



Overall, the model finds a nice balance between performance and interpretability. The monotonicity constraints help keep the model's behavior consistent with how real lenders think, and the threshold of 0.40 helps catch more Bad borrowers. The trade-off is that a few more Good borrowers may get rejected, but that's usually acceptable in a real credit-screening scenario.

2.4. Recommendations/Future Work

So far, we have achieved a stable model with effective metrics, from the perspective of the lender or lending organization. We understand that there is a trade-off between a few possible false negatives towards good borrowers being classified as bad borrowers, but with the volume of data we had and the quality that was provided, at present not many avenues seem possible for improvement given the business constraints of our problem statement. For expansion on this work, we would require more

volume of data with better initial quality (specifically fewer entries of borrowers having unverified or unavailable information), along with larger compute sources for possibly a wider hyperparameter search (a few parameters often chose their “best value” from their respective buckets at the ends of their respective ranges, implying better values might exist at the extremes).

3. Project Management

3.1. Final Schedule & Task Distribution (Gantt chart)

As the project developed, the final timeline changed, but the general framework remained the same. The team was kept organised and everyone was able to participate where they were most effective by assigning each section of work to the person whose talents fit the assignment.

Aditya Hegde guided much of the technical direction of the project across all stages. He was the one who looked for and evaluated the potential datasets early on, which in turn made the group choose the FICO dataset. He also took the initiative to set up the GitHub repository and draft the first report. In the midterm, he was deeply involved in the exploratory analysis, established an automatic data import system for the team, and experimented with early models like logistic regression and random forest to generate the slides. For the case study, he and other team members went through numerous papers, and he played a crucial role in deciding on the credit risk scoring paper which was explainable. He also wrote the solution part based on his close interaction with the text. In the last part, he worked mainly on creating and upgrading the XGBoost model, executing the data cleaning, tuning, threshold optimization, and monotonic constraints for a powerful as well as an understandable final solution.

Max Dokukin played a key role at every stage of the project. For the initial report, he created the shared Google Doc, led group discussions, completed the Goals and Objectives and Group Presentation sections, and helped with formatting and finalizing the document. During the midterm, he guided the team's discussions again, created a quarter of the slides, which included the project idea, introduction, and conclusion. He also reviewed and rehearsed with the group and researched the history of lending to improve the presentation. For the case study, he searched for relevant research papers, discussed them with the team, read the selected paper twice, created the shared document, and prepared the introduction and problem statement slides.

Charan Sai spent most of his time shaping the project's presentation. He ensured the story behind the solution was clear from start to finish. He wrote major sections of the report, improved the way the problem was introduced and explained, and helped make the team's technical work easy to understand. He reviewed everyone's contributions and kept the writing consistent. This way, the report felt unified rather than pieced together. He also suggested ideas for how the project could grow in the future and was actively involved in comparing the different models the team tested. His input was key to why the group confidently chose the final model.

Girish Manne was responsible for organizing the workflow and monitoring it. He regularly assessed the status of the tasks, modified the Gantt chart, and ensured that the project reached its milestones smoothly. He was actively participating in the discussions about the general solution and had a say in the choice of the modeling technique. He not only worked on planning but also created important visual aids for data analysis

and performance metrics that enhanced the understanding of the data for the team. Furthermore, he was involved in the assessment of different models and worked together with the team to pick the one that was most suitable for the project.

Task Distribution (Gantt Chart):

<https://docs.google.com/spreadsheets/d/17u5mn73ZLXXegFdsrXT6y0IMYG3wOpIkUPP7G426-iU/edit?gid=1788521638#gid=178852163>

3.2. Challenges

The project came with its own set of challenges that were tied directly to the work rather than logistics. One of the initial problems was to get a better understanding of the dataset more than expected. The data took more cleaning and restructuring computer time than initially estimated, and the team had to go back and question several assumptions regarding which features were indeed significant. This forced a couple of early modeling attempts to be reconsidered.

When the team started experimenting with various modelling techniques, another difficulty emerged. A range of choices were assessed as each member investigated theories that fit their personal logic. The measurements clarified the conclusion after the results were examined side by side. The group started using XGBoost as the main model since it constantly outperformed the other models. It required thorough study rather than preference to get there, and it enabled the team to rely on facts rather than conjecture.

There was also the challenge of keeping the documentation and the growing codebase in sync. With contributions coming in at different stages, the team had to be intentional about

checking for overlapping work and missing context. Over time this became easier as communication tightened and everyone used the shared tools more consistently. The repository and documentation gradually settled into a cleaner structure that supported the final stages of the project.

These challenges ended up helping the team rather than slowing it down. They encouraged a more deliberate way of working and a clearer sense of how each component of the project should fit together.

3.3. Learning Outcomes

We have acquired many significant takeaways through the project that were more than just completing the necessary work. A more practical understanding was developed by each member of how the strengths of individuals can be combined into a single workflow.

Aditya grew more confident in breaking down technical setups in a way that helped the rest of the group build on them without confusion, especially as he shaped the background and core environment for the project. Max figured out how to connect the dots between academic research and practical work by going through and analyzing the best case study papers and setting up a clean and well-organized repository that supported the team throughout the project. Charan enhanced his ability to tell a clear story about the technical work while at the same time putting forward creative ideas for future directions and assisting in comparing models to find the best one. Girish gained a better understanding of how planning tools direct real progress by keeping the Gantt chart updated, making useful visualizations like the correlation heatmap, and being part of the model evaluation that resulted in the final choice. Together these

experiences gave our whole team a clearer understanding of how individual strengths, when combined thoughtfully, can drive a project forward in a meaningful way.

Collectively, we learned how to evaluate ideas critically without dismissing them too early. The conversations about different modeling methods made it easier for people to question the basis of a problem, ask for clarification, and support their preferences with reasons rather than by simply stating their personal opinions. Furthermore, the whole experience proved that if people do really communicate their views and thoughts even when they are different, then a project becomes much smoother.

4. References

1. W. Wang, C. Lesner, A. Ran, M. Rukonic, J. Xue, and E. Shiu, "Using Small Business Banking Data for Explainable Credit Risk Scoring", *AAAI*, vol. 34, no. 08, pp. 13396-13401, Apr. 2020.